

Acid-Base Disorders — Stepwise ABG Interpretation

pH status → primary process → anion gap in EVERYONE (+ΔΔ if AGMA) → compensation · normal pH + abnormal values = mixed disorder

Normal values & definitions

Value	Normal	Derangement
pH	7.40 ± 0.05	<7.35 acidemia · >7.45 alkalemia
PCO ₂	40 ± 4	>44 resp acidosis · <36 resp alkalosis
HCO ₃	24 ± 2	<22 met acidosis · >26 met alkalosis
PO ₂	>80 RA	hypoxemia → check A–a gradient
Base excess	0 ± 2	>+2 met alkalosis · <-2 met acidosis

- **-emia** = blood pH; **-osis** = process — several -oses can coexist with any pH

Anion gap — calculate in every patient

- AG = Na – (Cl + HCO₃); "normal" 8–12 but **expected AG ≈ 2.5 × albumin** — hypoalbuminemia hides gaps
- An elevated AG is **never compensatory** — AGMA can hide behind any primary disorder
- **ΔΔ ("residual HCO₃")**: HCO₃ + (measured AG – expected AG) → ~24 pure AGMA · >26 + met alkalosis · <22 + NAGMA

Differentials

AGMA	MUDPILES : methanol, uremia, DKA/ketoacidosis, paraldehyde, INH/iron, lactic, ethylene glycol (+EtOH), salicylates
NAGMA	Gut HCO ₃ loss (diarrhea, ileostomy) · renal (RTA, early renal failure, ↓aldo) · excess Cl (NS). UAG: neGUTive = GI; positive = renal (unreliable in renal dysfxn)
Resp acidosis	↓elimination: obstructive/intrinsic lung dz, neuromuscular weakness, ↓drive (opioids/benzos) · ↑production: sepsis, overfeeding, hyperthyroid
Met alkalosis	Cl/IVF-responsive: vomiting/NGT, thiazide/loop, contraction · non-responsive: ↑aldo, genetic, milk-alkali, hypokalemia
Resp alkalosis	Anxiety/pain, fever/sepsis, CNS dz, salicylates, liver dz, PE

Compensation — PCO₂ & HCO₃ move the SAME direction

Primary	Expected compensation
Met acidosis	Winter's : PCO ₂ = 1.5×HCO ₃ + 8 (±2) ≈ thumb rule: PCO ₂ ≈ last 2 digits of pH
Met alkalosis	PCO ₂ = 0.7×HCO ₃ + 21
Resp acidosis	per ↑10 PCO ₂ : HCO ₃ +1 acute · +4 chronic (2–3 d)
Resp alkalosis	per ↓10 PCO ₂ : HCO ₃ –2 acute · –5 chronic

- Lungs compensate in **minutes**; kidneys in **2–3 days**
- Compensation **never overshoots** and never fully normalizes pH
- Measured ≠ expected = **second primary process** (PCO₂ > Winter's → + resp acidosis; < → + resp alkalosis)

Pattern grid (basic cases)

pH	PCO ₂	HCO ₃	Read
↓	nl/↓	↓	Metabolic acidosis (↓PCO ₂ = compensation)
↓	↑	nl/↑	Respiratory acidosis (↑HCO ₃ = compensation)
↓	↑	↓	Concurrent metabolic + respiratory acidosis
↑	nl/↑	↑	Metabolic alkalosis (↑PCO ₂ = compensation)
↑	↓	nl/↓	Respiratory alkalosis (↓HCO ₃ = compensation)
↑	↓	↑	Concurrent metabolic + respiratory alkalosis

Patterns are a screen — confirm with the compensation formulas; acute buffering alone shifts HCO₃ slightly.

Worked example (3 disorders in one gas)

- RYGB + pancreatitis + diarrhea + morphine: **6.98/41/58/10**; Na 146, Cl 121, HCO₃ 7, albumin 2.1, +ketones
- AG = 146 – (121 + 7) = 18; expected (2.5×2.1) ≈ 5–6 → **AGMA** (keto + lactic)
- ΔΔ: 7 + 12 = 19 < 22 → **+ NAGMA** (diarrhea)
- Winter's: 1.5×10+8 = 23 << measured 41 → **+ respiratory acidosis** (opioid)

Do / AVOID

- **DO** run all four steps on every gas — the order catches hidden disorders
- **DO** correct the expected AG for albumin before calling a gap "normal"
- **AVOID** calling a normal pH "no disorder" when PCO₂/HCO₃ are abnormal
- **AVOID** attributing an elevated AG to "compensation" — it never is
- **AVOID** the urinary AG in renal dysfunction